ANSWER KEY - The Kelly Capital Growth Investment Criterion - Full Book Quiz

Kelly Jr. 1956 / Thorp 1969/1971/2008 / MacLean-Thorp-Ziemba 2010 / Lv-Meister 2009 / Bell-Cover 1980

[Part I - The Kelly Formula & Information Theory]

1. Kelly (1956) defined exponential growth rate as G = lim(1/N)log(V_N/V_0).
2. The maximum growth rate G_max equals Shannon's channel capacity R.
3. For a coin toss (win prob p, loss prob q), the Kelly fraction f* = p - q.
4. The Kelly criterion maximizes the expected value of the logarithm of wealth each period.
5. For p=0.6, q=0.4, the optimal Kelly fraction f* = 0.20.

[Part II - Optimality Properties (Breiman / Thorp)]

6. Breiman (1961) proved Kelly asymptotically maximizes E of log X_N.
7. Kelly dominates all essentially different strategies almost surely.
8. Kelly reaches any asymptotic goal faster than any other essentially different strategy.
9. When G(f) < 0, wealth X_n converges to zero almost surely.
10. The critical fraction f_c satisfies G(f_c) = 0.

[Part III - Continuous Time & Portfolio Matrix]

11. Continuous time Kelly growth rate: g_inf(f) = r + f(m-r) - s^2 * f^2 / 2.
12. Continuous time optimal Kelly fraction: f* = (m-r)/s^2.
13. For multiple assets, optimal Kelly vector: F* = C^(-1) x [M-R].
14. Betting exactly double the Kelly amount yields growth equal to the risk-free rate.

[Part IV - Good and Bad Properties]

15. Good: The ElogX bettor has an optimal myopic policy - no need for past/future data.
16. Good: E(X/X*) <= 1 for all other portfolios X (Bell & Cover, 1980).
17. Good: No opponent can outperform Kelly wealth X* by factor t with prob > 1/t.
18. Bad: For n wins and n losses, Kelly wealth W_2n = W_0*(1-f^2)^n - bettor is behind.
19. Bad: Unweighted average return converges to half the arithmetic rate of return.

[Part V - Fractional Kelly & Utility]

20. Half Kelly corresponds to negative power utility d*w^d with d = d = -1.
21. Fractional Kelly alpha invests alpha in Kelly and 1 - alpha in cash.
22. Buffett appears to act as a full Kelly bettor subject to the no-borrowing constraint.
23. Keynes behaved like an 80% Kelly bettor with utility -W^(-0.25).

[Part VI - Error Sensitivity & Simulations]

24. Chopra & Ziemba (1993): errors in means vs. covariances are about 20 : 1 in importance.
25. At risk tolerance 50, mean:variance:covariance loss ratio = 22.50 : 10.98 : 2.05.
26. MacLean et al. simulation: 700 investment decisions, 2,000 scenarios, EV=1.14.
27. Full Kelly (f=1): mean final wealth approx $524,195 from initial $1,000.
28. Full Kelly minimum final wealth across 2,000 scenarios: $4.
29. Half Kelly minimum final wealth: $111 (vs $4 for full Kelly).

[Part VII - Thorp's Fund, Paradoxes & OU Processes]

30. Thorp's fund ran approx 28.5 years at ~20% annual compound return, ~6% std dev.
31. $10,000 in Thorp's fund grew to approx $18 million over that period.
32. Thorp coined the term Fortune's Formula for the Kelly criterion.
33. Proebsting's Paradox: successively better odds leads Kelly bettor to invest 100% of wealth.
34. For Ornstein-Uhlenbeck processes, optimal Kelly fraction: f* = R^(-1) * c_t.
35. Single stock OU optimal fraction: f* = (mu_t - r) / s^2.